XXXX: FULL STACK USING FLASK AND DJANGO

Objective: *To equip students with the knowledge and skills to develop, integrate, and deploy full stack web applications using Flask and Django frameworks.*

Prerequisite: Basic proficiency in Python programming, HTML/CSS, and foundational knowledge of web technologies and client-server architecture.

CREDIT-03 **L-T-P-J:3-0-0-0**

Module No.	Content	Teaching Hours
I	Introduction to Web Development, HTTP, MVC, REST Setup Flask, create first app, HTML, CSS, JavaScript Essentials, Build static web pages, Flask Routing, Templates (Jinja2), Forms, Create dynamic pages, Handling POST/GET Requests, Form Validation, Build Flask forms with validation, Flask-SQL Alchemy & Database Integration, CRUD operations with SQLite/PostgreSQL, User Authentication: Flask-Login, Flask-Bcrypt, Register/Login/Logout, Flask Blueprints and Modularization Refactor app into modules, REST API Development with Flask-RESTful, Create and test APIs, JSON Handling, Postman, Token Auth (JWT), Secure REST APIs, Deployment: GitHub, Docker, Heroku, Deploy Flask App to cloud, Mini Project Week, Full-featured Flask app with CRUD + Auth	21
II	Django Overview, MVT, URL Routing, Views, Create Django project, Templates, Static Files, Template Tags, Dynamic HTML rendering, Django Models, Admin Interface, Create models and use admin, Django Forms & Validation. Input and model forms, Django Authentication System, Login/Signup, password handling, Django REST Framework: Serializers, Views, Create APIs for models, Class-Based Views & ViewSets, Use generic views and routers, API Security: Permissions, JWT, Throttling, Secure your Django APIs, Frontend-Backend Integration (Axios/Fetch), Connect frontend with Django APIs, Advanced Topics: Pagination, Filtering, Signal, Optimize backend logic, Capstone Project Full Stack App Development (Frontend Django DB), Testing, Debugging, Deployment (Docker/Vercel), CI/CD & Dockerize Django, Final Project Presentation & Viva, Project Demo	21

Text Books:

- Miguel G., Flask Web Development: Developing Web Applications with Python, Publisher: O'Reilly Media, ISBN: 9781491991732
- William S. Vincent – Django for Beginners: Build websites with Python and Django, Publisher: WelcomeToCode, ISBN: 9781735467207
- William S. Vincent – Django for APIs: Build web APIs with Python and Django , Publisher: WelcomeToCode, ISBN: 9781735467221

Reference Books:

- Antonio Mele – Django 4 By Example: Build powerful and reliable Python web applications from scratch
- Publisher: Packt Publishing, ISBN: 9781801813051.
- Rick van Hattem – Mastering Python Web Development, Publisher: Packt Publishing,ISBN: 9781785886774.
- Julia Elman & Mark Lavin – Lightweight Django: Using REST, WebSockets, and Backbone.js, Publisher: O'Reilly Media, ISBN: 9781491945940

Outcome: After completion of Lab, the student will be able to:

- CO1: To provide a solid foundation in full-stack web development by introducing core web technologies, backend architectures (MVC/MVT), and development tools.
- CO2: To develop proficiency in building backend applications using Flask and Django frameworks.
- CO3: To enable the creation of RESTful APIs and their integration with frontend technologies.
- CO4: To implement database-driven web solutions using SQL and ORM tools like SQLAlchemy and Django ORM, emphasising CRUD operations and data modelling.
- CO5: To implement real-world deployment by covering containerization, version control, cloud deployment, and full-stack project development with testing and debugging.
- CO6:To cultivate teamwork and project management skills through capstone project development.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1/PSO3
CO2	PO2,PO3/PSO1
CO3	PO4/PSO1,PSO3
CO4	PO2,PO3/PSO4
CO5	PO2,PO4/PSO4
CO6	PO1/PSO2,PSO4

XXXX: FULL STACK USING FLASK AND DJANGO LAB

***Objective*:** *The objective of this course is that students will understand and implement full stack development with Django.*

CREDIT:01 **L-P-T-J:0-0-2-0**

Module No.	Content	Teaching Hours
I & II	• Build a simple Flask application with two routes and render content using HTML templates. • Create a contact form using Flask-WTF, validate inputs, and display submitted data. • Develop a basic CRUD (Create, Read, Update, Delete) app using Flask and SQLAlchemy. • Implement a user authentication system using Flask-Login with hashed passwords. • Develop a simple API to manage tasks or products and test it using Postman. • Set up a Django project and create a simple web app with views and templates. • Create models, register them in the admin panel, and perform CRUD via Django admin. • Create a REST API using Django REST Framework (DRF) with serializers and viewsets. • Develop a mini full-stack project (e.g., blog, task manager) using Flask or Django with frontend, database, authentication, and deployment.	20

Text Books:

- Miguel G., Flask Web Development: Developing Web Applications with Python, Publisher: O'Reilly Media, ISBN: 9781491991732
- William S. Vincent – Django for Beginners: Build websites with Python and Django, Publisher: WelcomeToCode, ISBN: 9781735467207
- William S. Vincent – Django for APIs: Build web APIs with Python and Django , Publisher: WelcomeToCode, ISBN: 9781735467221

Reference Books:

- Antonio Mele – Django 4 By Example: Build powerful and reliable Python web applications from scratch. Publisher: Packt Publishing, ISBN: 9781801813051.
- Rick van Hattem – Mastering Python Web Development, Publisher: Packt Publishing,ISBN: 9781785886774.
- Julia Elman & Mark Lavin – Lightweight Django: Using REST, WebSockets, and Backbone.js, Publisher: O'Reilly Media, ISBN: 9781491945940

Outcome: After completion of Lab, the student will be able to:

- CO1: To provide a solid foundation in full-stack web development by introducing core web technologies, backend architectures (MVC/MVT), and development tools.
- CO2: To develop proficiency in building backend applications using Flask and Django frameworks.
- CO3: To enable the creation of RESTful APIs and their integration with frontend technologies.
- CO4: To implement database-driven web solutions using SQL and ORM tools like SQLAlchemy and Django ORM, emphasising CRUD operations and data modelling.
- CO5: To implement real-world deployment by covering containerization, version control, cloud deployment, and full-stack project development with testing and debugging.
- CO6:To cultivate teamwork and project management skills through capstone project development.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs):

COs	POs/PSOs
CO1	PO1/PSO3
CO2	PO2,PO3/PSO1
CO3	PO4/PSO1,PSO3
CO4	PO2,PO3/PSO4
CO5	PO2,PO4/PSO4
CO6	PO1/PSO2,PSO4